Which Coefficients Need Correction? Design-Effect-Ratio Diagnostics and Selective Calibration for Bayesian Meta-Analysis With Dependent Effect Sizes

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Abstract. Bayesian meta-analysis with dependent effect sizes has recently acquired a robust reporting option: the posterior from a convenient working model can be calibrated so that its covariance matches a declared cluster-robust variance target. The correction is currently applied to the entire coefficient block at once, as if every coefficient required it to the same degree. This premise fails in a predictable way. Coefficients differ systematically in how much of the within-study dependence they inherit, and estimands identified by comparisons within studies are protected twice, by cancellation of shared sampling errors and by hierarchical shrinkage, so that blanket rescaling narrows exactly the intervals that were already wide enough. We propose the design-effect ratio, the ratio of the declared target variance to the working-posterior variance of a pre-specified estimand, as a two-sided diagnostic: estimands whose posterior understates the target are calibrated, and the remainder are left alone, never narrowed. Exact decompositions, transferred from Bayesian survey inference through a recently established likelihood correspondence, explain both regimes and show that coefficient-level ratios cannot stand in for contrast-level ratios. In two simulation studies with known truth and a re-analysis of a 117-study synthesis of brief alcohol interventions, the selective workflow holds near-nominal coverage where the frequentist benchmark undercovers (.975 against .904 for the weakest slope), avoids the coverage loss that blanket calibration incurs on protected estimands, and uses up to one fifth less interval width. We conclude that the decision to calibrate should be made estimand by estimand. An R package implements the workflow.

Date: August 2026.

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This work was supported by the Institute of Education Sciences, U.S. Department of Education, through Grant R305D240078 to the University of Alabama. The opinions expressed are those of the author and do not represent views of the Institute or the U.S. Department of Education.

Keywords: meta-analysis; dependent effect sizes; robust variance estimation; calibrated Bayes; design effect

1. Introduction

Meta-analyses routinely include several statistically dependent effect size estimates from each study, for instance when trials measure several outcomes or several follow-up occasions. A common strategy for handling such dependence is to fit a tractable working model and to report uncertainty using a cluster-robust variance estimator. Hedges et al. (2010) introduced this approach for meta-regression, and Tipton (2015), Tipton and Pustejovsky (2015) and Pustejovsky and Tipton (2018) developed the small-sample corrections, based on the bias-reduced linearization of Bell and McCaffrey (2002) together with Satterthwaite reference distributions, that have since become standard practice (for practical guidance see Tanner-Smith et al., 2016; Tipton et al., 2019); Pustejovsky and Tipton (2022) extended the range of working models.

A Bayesian version of this strategy has recently become available. An analyst can fit the same working models by MCMC, but the covariance of the resulting posterior reflects the prior and the assumed working correlation rather than any robust target, so a posterior interval is not a robust interval. Lee, Tipton, et al. (2026) identified a marginal fixed-effect sandwich covariance as the appropriate reporting target for the Bayesian analysis, and Lee and Tipton (2026) proposed a center-preserving Cholesky transformation that calibrates the posterior draws to this target exactly, with CR2 small-sample corrections and per-coefficient Satterthwaite reference distributions. In a 324-condition simulation and a re-analysis of a large synthesis of brief alcohol interventions, the calibrated intervals attained near-nominal frequentist coverage in conditions where the frequentist standard of care did not.

In this paper we consider a question that this line of work leaves open: should the calibration be applied to all estimands alike? At present the correction is applied to the entire coefficient block at once. The premise is that every coefficient understates the declared target, and to roughly the same degree. Neither part is correct. Coefficients differ systematically in how much of the within-study dependence they inherit, a fact that can be established analytically in the hierarchical Gaussian model and that is visible in applied data. More troublesome, the blanket transformation narrows the posterior for estimands whose working posterior is already wide relative to the target.

In the simulations of Section 6, blanket calibration covers the within-study moderator coefficients less well than the uncorrected posterior does: the correction does its damage exactly where the hierarchical model was already performing well.

To discriminate among estimands we propose a diagnostic that we call the *design-effect ratio* (DER). For each pre-specified coefficient or scientific contrast, the DER compares the variance that the declared robust target assigns to the estimand with the variance that the working posterior assigns to it; the formal definition is given in Section 2. The diagnostic is two-sided. A ratio well above one identifies estimands whose posterior understates the declared target, and these are calibrated. A ratio at or below one identifies estimands for which the correction is unnecessary and, if applied automatically, harmful, and these are left alone; in particular they are never narrowed. The workflow that results is selective: the analyst registers the estimands in advance, computes the ratio for each, calibrates the flagged set, and leaves the remainder of the posterior untouched.

The diagnostic has a mechanism behind it, not only a motivation. In the hierarchical Gaussian model, the dependence-layer ratio of any estimand decomposes exactly into a within-study part and a between-study part (Section 4). Estimands identified by comparisons within studies are protected twice, once because positively correlated sampling errors partially cancel in such comparisons, and once by hierarchical shrinkage; estimands identified by variation between studies inherit design-effect factors attenuated by shrinkage. The weight that mixes the two parts is computable from the design alone, and it explains why a contrast can be protected while both coefficients it involves are exposed: coefficient-level ratios do not determine contrast-level ratios, and Section 3 shows that no function of them does.

The name is retained deliberately. Design-effect ratios were developed as parameter-level diagnostics for selective design-based correction in Bayesian survey inference (Lee, Williams, & Savitsky, 2026), where the correction being gated is the pseudo-posterior rescaling of Williams and Savitsky (2021). Since the correlated-effects working model of dependent-effects meta-analysis and the unit-weighted clustered Bayesian survey model assign identical likelihoods (Lee, Tipton, et al., 2026), the diagnostic transfers between the two settings, and the survey-side decomposition theorems transfer with it. The meta-analytic setting in fact sharpens them: the within-study dependence here is an explicit equicorrelation structure rather than a scalar parameterization, so the transferred theorems resolve the direction of an estimand in a way the survey formulation cannot.

Some limits of scope should be understood from the start. We prove no coverage theorem: the DER diagnoses, and the selective correction calibrates to, a named variance target, and frequentist coverage is examined by simulation under stated data-generating processes rather than established in general. Nor does the diagnostic certify unflagged estimands: a ratio at or below one is a statement about dispersion relative to the declared target, not a statement about bias, and it carries no guarantee. Heterogeneity parameters are outside its reach altogether, in meta-analysis as in the survey setting (Lee, Williams, & Savitsky, 2026).

In frequentist meta-analysis, analysts often compare a model-based and a robust standard error informally, but the comparison carries no rule for acting on it. A formal selective-correction framework with decomposition results exists in the survey setting (Lee, Williams, & Savitsky, 2026); as far as we are aware, no estimand-level gating of a posterior calibration has been proposed for meta-analysis, where the absence of robust uncertainty methods for dependent effects has itself been noted as a gap (Bartoš et al., 2026). The closest existing approaches calibrate composite-likelihood posteriors to Godambe targets (Wang et al., 2025) or adjust posteriors using sandwich covariance machinery in general models (Li & Rice, 2024; Müller, 2013; Shaby, 2014), but none of these gates the adjustment estimand by estimand. The rationale for combining frequentist targets with Bayesian reports comes from the calibrated-Bayes tradition (Little, 2004, 2012; Rubin, 1984), and the design-effect vocabulary descends from Kish (1965) and its estimand-specific generalization by Skinner (1989).

The plan of the paper is as follows. In Section 2 we describe the working models, define the declared variance target and the estimand-level DER, and introduce the estimand registry. Section 3 shows why coordinate-level ratios cannot substitute for estimand-level ratios. In Section 4 we derive exact decompositions for fixed and random effects, including a directional decomposition, an exact mixture representation, and a conservation identity, and we discuss their boundary behavior. Section 5 assembles these components into a selective compute–classify–correct workflow with per-estimand Satterthwaite reference distributions and rules for numerically fragile estimands. In Section 6 we evaluate the workflow in two simulation studies with known truth: a re-analysis of the stored replications of Lee and Tipton (2026) that requires no new posterior sampling, and a new study of registered scientific contrasts. Section 7 applies the workflow to the 117-study synthesis of brief alcohol interventions of Tanner-Smith and Lipsey (2015). Finally, we close the paper with a discussion in Section 8. The methods are implemented in the R package `bayesRVE` (Lee, 2026).

2. **Working models, declared targets, and the design-effect ratio**

2.1. **Data structure and working models.** Study $j = 1, \ldots, J$ contributes n_j effect size estimates y_{ij} with known sampling variances v_{ij}; we write $\bar{s}_j^2 = n_j^{-1} \sum_i v_{ij}$ for the study average and $N = \sum_j n_j$ for the total number of estimates. A K-vector of moderators x_{ij} is observed for each estimate. The correlated-effects (CE) working model includes a study-level random effect,

$$y_{ij} = x_{ij}^\top \beta + \theta_j + \varepsilon_{ij}, \qquad \theta_j \sim N(0, \tau^2),$$

with working within-study independence given θ_j at scale $\bar{s}_j^2$. The correlated-and-hierarchical-effects (CHE) model adds a within-study heterogeneity component ω^2. These are the working models of Pustejovsky and Tipton (2022), fit here by MCMC with weakly informative priors. We adopt the models, their Stan implementation, and the three-layer calibrated reporting procedure built on them from Lee and Tipton (2026), and we do not restate that machinery. The true sampling errors are dependent within studies, and under the equicorrelation model their correlation is ρ, a quantity that primary studies rarely report.

2.2. **The declared target.** Every diagnostic in this paper is a comparison against a declared variance target, and the choice of target has substantive consequences. The target that we use operationally for gating is the marginal cluster-robust sandwich for β with CR2 small-sample correction, constructed at a declared working correlation ρ_{work} and at plug-in variance components. This is exactly the object that the calibration of Lee and Tipton (2026) matches, so that diagnosis and correction share a single target. Declaring the target requires declaring everything the target conditions on: the aggregation convention (the working likelihood uses study-average variances, whereas the sandwich weights use the individual v_{ij}), the value of ρ_{work}, the small-sample correction, and the plug-in values. As Section 7 shows, these components move the diagnostic materially, so that an analyst who declares a different target is addressing a different, though equally legitimate, question. A theorem-level conditional target, which isolates the dependence mechanism alone, is defined in Section 4; it is used for interpretation but never for gating.

2.3. **The design-effect ratio.**

Definition 1 (Estimand-level DER)**.** Let $\Sigma = \mathrm{Cov}_{\text{post}}(\beta \mid y)$ be the working-model posterior covariance of the fixed effects and V_T a declared target covariance for the same vector. For a nonzero estimable contrast $c^\top\beta$ with $c^\top\Sigma c > 0$,

$$\mathrm{DER}_T(c^\top\beta) = \frac{c^\top V_T\, c}{c^\top\Sigma\, c}, \qquad \text{with standard-error factor } \sqrt{\mathrm{DER}_T(c^\top\beta)}.$$

Diagonal coefficient DERs are the special case $c = e_k$. A value near one indicates that the working posterior already matches the declared target; a value above one indicates that the posterior understates the target, so that calibration widens the interval; and a value below one indicates that the posterior is wider than the target. For reasons that Section 4 makes precise, the appropriate response in the last case is to leave the interval unchanged rather than to narrow it.

For a flagged *scalar* estimand, calibration is a one-dimensional rescaling of its own posterior draws,

$$(c^\top\beta)^{*(s)} \;=\; \overline{c^\top\beta} \;+\; \sqrt{\mathrm{DER}_T(c^\top\beta)}\,\bigl(c^\top\beta^{(s)} - \overline{c^\top\beta}\bigr), \tag{2.1}$$

which preserves the posterior mean exactly and matches the target variance exactly in one dimension. Registered contrasts therefore never require the multi-coefficient machinery of Section 5: the analyst gates on the contrast's own DER and rescales its own scalar posterior.

Throughout, ρ_{work} is a declared analytic convention and not an estimate of the true within-study dependence. Table 1 summarizes the objects that recur in the sections that follow. Since the DER is a property of a (target, estimand) pair and not of a model alone, we organize the analysis around an estimand registry: every DER reported in this paper corresponds to a pre-specified row naming the contrast vector, the target, the variant, and the intended action. Table 2 presents the registry for the applied analysis of Section 7.

3. Coordinate-level and estimand-level diagnostics

We define the DER at the level of estimands because coordinate-level ratios do not determine it; this section establishes that claim.

Note first that the ratio is invariant for estimands but not for coordinates. If the coefficient vector is reparameterized as $\beta = A\alpha$ with A nonsingular, then scores transform by $A^\top$, so that precision-type objects transform as $\Sigma_\alpha^{-1} = A^\top\Sigma_\beta^{-1}A$, and likewise for the target. The same scientific estimand, written $c^\top\beta = d^\top\alpha$ with $d = A^\top c$, then satisfies $\mathrm{DER}_T(d^\top\alpha) = \mathrm{DER}_T(c^\top\beta)$; that is, the contrast-level ratio is invariant for a fixed linear estimand under consistent transformation. It does not follow that diagonal DERs are invariant to the coding of the model, because after

TABLE 1. Recurring objects. The workflow gates on the operational target; theorem-layer objects serve interpretation.

Symbol	Object	Role
Σ	working-posterior covariance of β	DER denominator (the report being diagnosed)
V_T	operational target: CR2-corrected marginal sandwich at declared ρ_{work} and plug-ins	DER numerator; the gate acts on this quantity
$\mathbf{M}_{\text{cond}}, \mathbf{M}_{\text{marg}}$	theorem-layer meats (study-average scales; Section 4)	mechanism interpretation only
ext. benchmark	clubSandwich CR2 on a REML refit	cross-implementation check
$\omega_B(c)$	between-information weight of an estimand	explains the position of a DER
$\text{df}(c)$	per-estimand Satterthwaite degrees of freedom	reference distribution; gate-stability signal
c_0	gate threshold (default 1.2; varied in sensitivity analyses)	classification

TABLE 2. Estimand registry for the TSL15 audit (Section 7). All rows use the declared operational target (CR2-corrected marginal sandwich, $\rho_{\text{work}} = 0.6$); theorem-level values enter only as interpretation columns.

ID	Estimand	Contrast	Action rule
E01–E06	six outcome-category cell means	$e_1, \dots, e_6$	tier table (Section 5)
E07–E11	five moderators (follow-up week, long follow-up, % college, % male, attrition)	$e_7, \dots, e_{11}$	tier table
C1	heavy use − frequency of use	$e_2 - e_1$	gate, then rescale (2.1)
C2	grand outcome mean	$\frac{1}{6}\sum_{k=1}^{6} e_k$	gate, then rescale
C3	quantity of use − blood alcohol concentration	$e_3 - e_5$	gate, then rescale
θ_j	117 study effects (CE fit)	—	conditional widening (Section 4)

recoding the coordinate vector e_k names a different estimand. Cell-means coding, reference-cell coding and centered coding all give the same DER for the estimand "heavy use minus frequency of use" but different DERs for their respective coordinates.

The off-diagonal elements of both covariance matrices also enter the estimand-level ratio. For a two-cell comparison $\delta = \mu_2 - \mu_1$,

$$\text{DER}_T(\delta) = \frac{[V_T]_{11} + [V_T]_{22} - 2[V_T]_{12}}{\Sigma_{11} + \Sigma_{22} - 2\Sigma_{12}},$$

TABLE 3. Diagonal DERs cannot substitute for estimand DERs: four counterexamples. Σ = posterior covariance, V_T = target; equicorr(r) denotes unit variances with common correlation r.

Σ	V_T	Estimand	Diagonal DERs	Estimand DER
I_2	$\begin{pmatrix} 4 & 3.5 \\ 3.5 & 4 \end{pmatrix}$	$A - B$	4, 4	0.5
equicorr(0.8), K=2	I_2	$A - B$	1, 1	5
equicorr(0.8), K=4	I_4	DID $(1, -1, -1, 1)$	all 1	5
equicorr(0.8), K=4	I_4	mean $(1, 1, 1, 1)/4$	all 1	0.29

and the cross terms can reverse any conclusion drawn from the two diagonal ratios alone. Table 3 gives four covariance pairs, each verifiable in two lines of arithmetic, in which the diagonal reading fails in a different direction: the cell means can appear severely under-dispersed while their difference is over-dispersed, the cell means can appear calibrated while the difference is severely under-dispersed, and an interaction contrast and an average over the same four cells can fall at opposite ends of the scale.

The same phenomenon arises in applied syntheses. In the TSL15 audit of Section 7, the two outcome-category cell means involved in the registered comparison C1 carry empirical DERs of 1.09 and 1.25, so that one is flagged and the other is not, while the comparison itself carries a DER of 1.39 under the declared target and 0.41 under the theorem-level marginal target. No function of the two diagonal values predicts either quantity. These considerations lead to the registry rule adopted throughout: diagonal DERs are descriptive diagnostics for coordinate estimands, whereas planned scientific estimands require their own DERs computed from the full covariance matrices. The workflow of Section 5 accordingly takes the estimand registry, rather than the coordinate list, as its unit of analysis.

4. Exact decompositions for fixed and random effects

The results in this section concern the dependence mechanism in isolation. They are stated for the CE working model with known study-average scales, fixed $\tau^2 > 0$, a flat (or negligible-precision) prior on β, true within-study equicorrelation ρ, and a full-rank design. We refer to these as conditions M1–M5. Online Supplement S-A states them formally and maps them onto the survey-side conditions of Lee, Williams, and Savitsky (2026); the scalar design-effect parameterization of the survey-side theorems is not translated, because the equicorrelation structure supplies the meat directly. Under these conditions, the posterior covariance of β is exactly $\mathbf{S}^{-1}$, where $\mathbf{S}$ is the profiled precision matrix defined below, so that population DERs are ratios of computable matrices. Proofs appear in Online Supplement S-A, and every displayed identity in this section has been verified to machine precision against direct linear-algebra computation and Monte Carlo simulation (Supplement S-G).

4.1. **Fixed effects: a directional decomposition.** We write $a_j = n_j/\bar{s}_j^2$ and $B_j = \tau^2/(\tau^2 + \bar{s}_j^2/n_j)$ for the study precision and shrinkage factor, $\bar{x}_j$ for the study-mean moderator vector, W_j for the within-study centered scatter at scale $\bar{s}_j^{-2}$, and $\mathrm{DEFF}_j = 1 + (n_j - 1)\rho$ for the classical between-direction design effect (Kish, 1965).

Theorem 1 (Directional decomposition)**.** *Under M1–M5, the profiled precision and the conditional-target meat decompose as*

$$\mathbf{S} = \sum_j \big\{ W_j + a_j(1 - B_j)\, \bar{x}_j \bar{x}_j^\top \big\},$$

$$\mathbf{M}_{\text{cond}} = \sum_j \big\{ (1-\rho)\, W_j + \mathrm{DEFF}_j (1 - B_j)^2 a_j\, \bar{x}_j \bar{x}_j^\top \big\},$$

and $\mathrm{DER}_{\text{cond}}(c^\top \beta) = c^\top \mathbf{S}^{-1} \mathbf{M}_{\text{cond}} \mathbf{S}^{-1} c \,/\, c^\top \mathbf{S}^{-1} c$*. In particular: (a) for the intercept or any study-level moderator direction in the balanced case,* $\mathrm{DER}_{\text{cond}} = \mathrm{DEFF}\,(1 - B)$*, and the total-variance (marginal-meat) version equals* $B + \mathrm{DEFF}(1 - B)$*; (b) if* $\bar{x}_j^\top \mathbf{S}^{-1} c = 0$ *for all* j*, as holds for a within-study-centered covariate that is orthogonal within studies to the other moderators, then* $\mathrm{DER}_{\text{cond}}(c^\top \beta) = 1 - \rho$*.*

Part (b) describes a feature of the meta-analytic setting that does not arise in the survey formulation. Positively correlated sampling errors partially cancel in within-study comparisons, so a working-independence posterior overstates their variance. Purely within-study-identified estimands are thus protected twice, once by this cancellation, since $1 - \rho < 1$, and once by hierarchical shrinkage.

It is worth spelling out how this squares with the survey-side theory. The survey-side fixed-effect theorem (Lee, Williams, & Savitsky, 2026) parameterizes the within-cluster meat by a single score-weighted design effect, and therefore places the same within-cluster factor on every exposed direction; the survey conclusion is that within-cluster covariates are fully exposed. There is no contradiction, because the two settings place different covariance operators behind the same word "within". Under the scalar parameterization the design inflates score variance in every direction of a sampled cluster, whereas the meta-analytic dependence operator $C_n(\rho)$ has two eigenvalues, and within-study contrasts lie in the direction associated with the smaller one. The qualitative conclusion changes because the covariance operator changes.

Theorem 2 (Exact mixture representation). *Under M1–M5, let $u = \mathbf{S}^{-1}c$, and define the between weight and shrinkage-tilted between design effect*

$$\omega_B(c) = \frac{\sum_j (1-B_j)a_j(\bar{x}_j^\top u)^2}{u^\top \mathbf{S} u}, \qquad \overline{\mathrm{DEFF}}_B(c) = \frac{\sum_j \mathrm{DEFF}_j(1-B_j)^2 a_j(\bar{x}_j^\top u)^2}{\sum_j (1-B_j)a_j(\bar{x}_j^\top u)^2}.$$

Then, exactly,

$$\mathrm{DER}_{\mathrm{cond}}(c^\top \beta) \ = \ (1-\rho)\left(1-\omega_B(c)\right) \ + \ \overline{\mathrm{DEFF}}_B(c)\, \omega_B(c),$$

with the convention that if the between component vanishes, that is if $\sum_j (1-B_j)a_j(\bar{x}_j^\top u)^2 = 0$ as in the pure within-identified case, then $\omega_B(c) = 0$, the between term is absent, and $\overline{\mathrm{DEFF}}_B(c)$ (whose denominator is then zero) is immaterial. Moreover the between-information fraction $R(c) = 1 - (c^\top \mathbf{H}_{\beta\beta}^{-1} c)/(c^\top \mathbf{S}^{-1} c)$, with $\mathbf{H}_{\beta\beta}$ the θ-conditional precision, satisfies $0 \leq R(c) \leq \max_j B_j$.

It follows from Theorem 2 that the conditional DER of every estimand is an exact convex combination of the within design effect $(1-\rho)$ and a weighted average of shrinkage-attenuated between design-effect factors, with a weight $\omega_B(c)$ that can be computed from the design alone. The representation accounts directly for coordinate-versus-contrast reversals, since a difference of two cell means can have $\omega_B \approx 0$ while each cell mean has $\omega_B \approx 0.8$, the between-study loadings cancelling in the difference. It also replaces the informal question of whether a moderator is between-study or within-study by a numerical quantity, which we report alongside every DER in Section 7.

4.2. **Random effects, conservation, and the boundary.**

Theorem 3 (Random effects). *Under M1–M5, the conditional random-effect ratio is $\mathrm{DER}_{\theta_j}^{\mathrm{cond}} = B_j\, \mathrm{DEFF}_j$ for every design. In the balanced intercept-only model the exact finite-J version multiplies by $\kappa_j(J) = 1 - \{J(1-B_j) + B_j\}^{-1}$.*

Strongly shrunken study effects (small B_j) are protected. In contrast, large and precisely estimated studies with many effect sizes (B_j near one, large DEFF_j) have forest-plot intervals that understate the declared conditional target by a factor approaching DEFF_j. The practical implication is a per-study widening factor of $\sqrt{B_j \mathrm{DEFF}_j}$ for CE forest plots, which we apply in Section 7.

Corollary 1 (Conservation and boundary). *In the balanced intercept-only model, $\mathrm{DER}_\mu^{\mathrm{cond}} + \mathrm{DER}_\theta^{\mathrm{cond}} = \mathrm{DEFF}(1-B) + B\,\mathrm{DEFF} = \mathrm{DEFF}$: shrinkage reallocates design sensitivity between levels but does not remove it. In unbalanced data the analogous display is a* signed diagnostic, *with gap $\overline{B_j \mathrm{DEFF}_j} - \bar{B}\,\overline{\mathrm{DEFF}} = \widehat{\mathrm{Cov}}(B_j, \mathrm{DEFF}_j)$,* *typically positive because both factors increase with n_j. At the boundary $\rho = 0$,* $\mathrm{DER}_{\mu}^{\mathrm{cond}} = 1 - B \neq 1$ *while the total-variance ratio equals $B + (1 - B) = 1$.*

The boundary case in Corollary 1 motivates a central rule of the workflow. A conditional DER below one neither certifies over-coverage nor justifies narrowing the interval, since it typically indicates that shrinkage more than compensates for a small design effect, and it is the total-variance benchmark $B + \mathrm{DEFF}(1 - B)$ that remains at or above one. The workflow therefore never narrows an interval on the basis of a low DER, a rule that we refer to as the do-not-narrow rule. As Section 6 shows, this rule has measurable consequences, since blanket calibration, which does narrow such intervals, incurs a corresponding cost in coverage.

4.3. **The CHE model and the empirical target.** All results in this section transfer to the CHE working model under a two-parameter substitution, $\bar{s}_j^2 \mapsto \bar{s}_j^2 + \omega^2$ *and* $\rho \mapsto \rho_j^{\mathrm{che}} = \rho\, \bar{s}_j^2 / (\bar{s}_j^2 + \omega^2)$, conditional on plug-in ω^2, so that modeled within-study heterogeneity dilutes the effective error correlation, study by study (Supplement S-B). Because the CHE implementation integrates the study effects out analytically, only β-block DERs are defined under CHE, and we make no random-effect claims for that model.

The theorem-level conditional target defined above serves interpretation only. The operational target of Definition 1 additionally reflects the declared weighting convention (individual v_{ij} in the sandwich, study averages in the likelihood), the CR2 correction, and the plug-in components. Section 7 reports both layers side by side, and a stepwise decomposition in Supplement S-E attributes each large difference between the theorem-level and operational values to a specific declared component, principally the individual-variance weighting and plug-in step, with the CR2 correction contributing a smaller additional adjustment. In this way the effect of the declaration on the diagnostic is quantified rather than merely asserted. Whether gating on the operational target produces well-calibrated inference is an empirical question, which we examine against known truth in Section 6.

5. The selective workflow

The diagnostics of the previous sections lead to a reporting procedure with four steps, summarized in Figure 1.

Step 1: Register. The analyst pre-specifies the estimands: the coefficients that are themselves scientific quantities, together with every planned contrast, each with

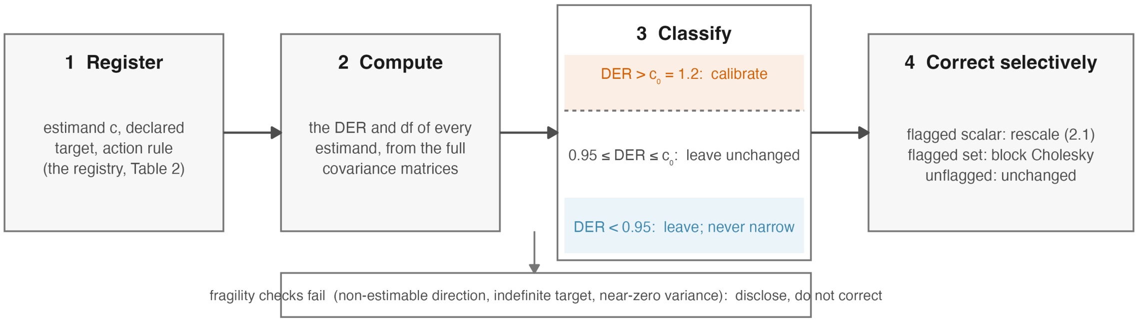

FIGURE 1. The selective workflow. Estimands are registered before analysis; the DER and its degrees of freedom are computed from the full covariance matrices; the classification is two-sided, with a calibrate zone above the gate $c_0 = 1.2$ and a do-not-narrow zone below 0.95 (the fine-grained tiers are in Table 4); flagged estimands are corrected by their own scalar rule or, for a jointly reported set, by the block transformation, while estimands that fail the fragility checks are disclosed rather than corrected.

its contrast vector, its declared target and its action rule (Table 2). No DER is interpreted without a corresponding registry row.

Step 2: Compute. The analyst fits the working model, constructs the declared target V_T (here the CR2-corrected marginal sandwich at declared ρ_{work} and posterior-mean plug-ins, which is the same object that the blanket calibration matches) and computes DER_T for every registered estimand from the full covariance matrices. Alongside each DER we compute a per-estimand Satterthwaite degree of freedom $\text{df}(c)$, which generalizes the per-coefficient construction of Tipton (2015) from e_k to arbitrary c through the same two-moment approximation (Supplement S-C); at $c = e_k$ this construction reproduces the per-coefficient values exactly.

Step 3: Classify. Each DER is mapped to an action tier (Table 4). Note that the two ends of the scale play different roles. The bottom tier implements the boundary analysis of Section 4: a low DER indicates that shrinkage more than compensates for the design effect relative to this target, and the workflow prohibits narrowing in this case. Above the gate threshold c_0 the estimand is calibrated. We adopt $c_0 = 1.2$, the conservative default of the survey setting (Lee, Williams, & Savitsky, 2026); Section 6 examines the range $c_0 \in [1.05, 2.0]$ and finds that coverage moves by less than one percentage point across it, so the specific choice of threshold is not critical. Because the estimated DER is itself subject to sampling variability where degrees of freedom are scarce, the gate is one-sided, in the sense that it can only widen intervals, and

TABLE 4. Action tiers. The gate $c_0 = 1.2$ adopts the conservative default of the survey framework (Lee, Williams, & Savitsky, 2026); the surrounding cutpoints are operational conventions of this paper, and Section 6 shows that operating characteristics move by less than one percentage point over $c_0 \in [1.05, 2.0]$.

Tier	DER_T	Reading	Action
T0	< 0.95	posterior wider than target	report; investigate if extreme; *never narrow*
T1	$[0.95, 1.05]$	calibrated to target	none
T2	$(1.05, 2]$	mild understatement	gate at c_0; calibrate if flagged
T3	$(2, 4]$	moderate understatement	calibrate; examine the estimand's identification
T4	> 4	severe understatement	calibrate; revisit specification and fragility checks

carries a stability label: when $\mathrm{df}(c) < 6$ or $|\log(\mathrm{DER}/c_0)| < \log 1.25$, the workflow marks the gate as unstable and reports the decision together with the full c_0 sensitivity analysis for that estimand rather than as a single verdict. Estimands that fail the numerical-fragility checks (a non-estimable direction, an indefinite target on the contrast, near-zero posterior variance, an inadequate effective number of draws, or, as operationalized for the simulation analyses, $\mathrm{DER} > 50$ or $\mathrm{df}(c) < 2$) constitute a third workflow outcome: they are disclosed, neither corrected nor certified (Supplement S-C).

Step 4: Correct selectively. For a flagged scalar estimand, including any registered contrast, the analyst applies the one-dimensional rescaling (2.1) and reports the interval $\overline{c^\top \beta} \pm t_{0.975,\,\mathrm{df}(c)}\,\mathrm{sd}(c^\top \beta^*)$. For a flagged coefficient *set* F that is to be reported jointly, the analyst applies the block-Cholesky calibration of Lee and Tipton (2026) restricted to F: with $L_1 L_1^\top = \Sigma_{FF}$ and $L_2 L_2^\top = [V_T]_{FF}$, the F-coordinates of each draw are transformed by $A_F = L_2 L_1^{-1}$ about the posterior mean, and the complement is left unchanged. Center preservation and exact F-block covariance matching carry over from the blanket case, restricted to F, and the computational cost is $O(|F|^3)$.

What selective correction does not guarantee matters as much as what it does. Cross-block covariances between flagged and unflagged coordinates carry no matching guarantee, so joint credible regions that require off-diagonal elements between the two groups should use the blanket transformation. Selective calibration is a tool for per-estimand marginal reporting. A further caution applies within the block. The block transformation matches the target on the flagged coordinates, but it also alters every linear combination of those coordinates, including directions that were not individually flagged. A registered contrast should therefore always be reported through its own scalar rule (2.1) rather than computed from block-transformed coefficient draws. Registered contrasts avoid the cross-block issue entirely, because (2.1) calibrates the estimand's own one-dimensional posterior. Finally, no step of the workflow certifies frequentist coverage. Whether the procedure produces well-calibrated inference is an empirical question, which we examine in the following section.

6. **Simulation studies**

We evaluate the workflow against known truth in two complementary simulation studies. Both are anchored to the same empirical database: study structures, moderator values and sampling-variance profiles are resampled from the Tanner-Smith–Lipsey synthesis (Pustejovsky & Tipton, 2022; Tanner-Smith & Lipsey, 2015), and the true coefficients are fixed at values estimated from that database.

6.1. **Coefficient level: a re-analysis with no new sampling.**

The 324-condition simulation of Lee and Tipton (2026) stored, for every replication and coefficient, the calibrated and uncalibrated interval endpoints together with the per-coefficient calibration ratio, which, by the exact-matching property of the calibration, equals $\sqrt{\mathrm{DER}}$ against the CR2 target. We verified this identity on 1.9 million stored rows before use, and the two computational routes agree to a median relative deviation of 3×10^{-16} (Supplement S-D). Because the uncalibrated and calibrated methods share each Stan fit, the DER-gated selective estimator can be synthesized retroactively, replication by replication: the calibrated interval is used when the stored DER exceeds c_0, and the raw posterior interval is used otherwise. Note that the gate uses only quantities that an analyst would possess in practice. We evaluate the correctly specified arms within the operating envelope ($J \geq \max(8, 2K + 2)$), restricted to replications in which all comparators converged, giving 955,762 coefficient-level replications that span $J \in \{10, \ldots, 80\}$, $K \in \{1, 4, 7\}$, $\rho \in \{0, 0.3, 0.6, 0.8\}$ and two heterogeneity structures.

Table 5 and Figure 2 summarize the results; all statements refer to the conditions studied. Where correction is needed, gating sacrifices very little: on the slope coefficients the selective estimator covers within half a percentage point of blanket calibration (.970–.975 versus .966–.978) while using 79–93% of blanket calibration's width on the coefficients where the two differ most. Where the working posterior is already wide relative to the target, gating does better than blanket calibration rather

TABLE 5. Coefficient-level operating characteristics (nominal 95%; correct specification, in-envelope, jointly converged; unweighted means over conditions). Monte Carlo standard errors $\leq$ 0.001; between-condition standard errors shown in Figure 2. Width = median interval width; share = fraction of replications gated into correction at $c_0 = 1.2$.

	Coverage					width ratio (sel./blanket)
Coefficient	raw	selective	blanket	CR2	share	
β_0 (average effect)	.953	.954	.952	.948	.03	1.00
β_1 (small slope)	.962	.975	.978	.904	.36	0.79
β_2 (larger slope)	.961	.970	.971	.961	.27	0.93
β_3 (null slope)	.965	.971	.966	.956	.18	0.99
β_4–β_6 (within dummies)	.956–.965	.964–.972	.952–.961	.952–.961	.18	≈1.0

than worse: on the within-study dummy coefficients, whose median DER falls below one (Figure 3), blanket calibration narrows the intervals and covers at .952–.961, whereas the do-not-narrow gate maintains .964–.972. This is the consequence of indiscriminate correction that the boundary analysis of Corollary 1 predicts. Against the frequentist benchmark the comparison runs the other way: on the small slope, where CR2 is known to have difficulties, CR2 covers at .904 and the gated estimator at .975. The costs are those documented by Lee and Tipton (2026): power on the small slope is low (.030 gated and .027 blanket, against .101 for CR2), and Type-I error on the null slope is controlled for all Bayesian variants (.029–.035, against .044 for CR2).

The estimated DER is itself subject to sampling variability where degrees of freedom are scarce, and the gate inherits this variability. In the fully correctly specified null case ($\rho = 0$, no within-study heterogeneity), the gate still fires on 27.5% of slope replications (3.5% for the average effect). These are false positives. Their cost is a mildly conservative interval rather than an invalid one, but they should be acknowledged, and the unstable-gate label of Section 5 is designed for such cases.

Two concerns about the gate itself can be examined in the same stored data. The workflow does not depend on a finely tuned cutoff: selective slope coverage moves only from .977 to .969 as c_0 ranges over $[1.05, 2.0]$. A more delicate concern is that gating on an estimated quantity could induce selection effects. The conditional-coverage diagnostics of Supplement S-D address this directly: coverage of the gated report is 0.995 among flagged and 0.959 among unflagged replications, 0.970 in the band nearest the threshold, and highest rather than lowest in the lowest degrees-of-freedom

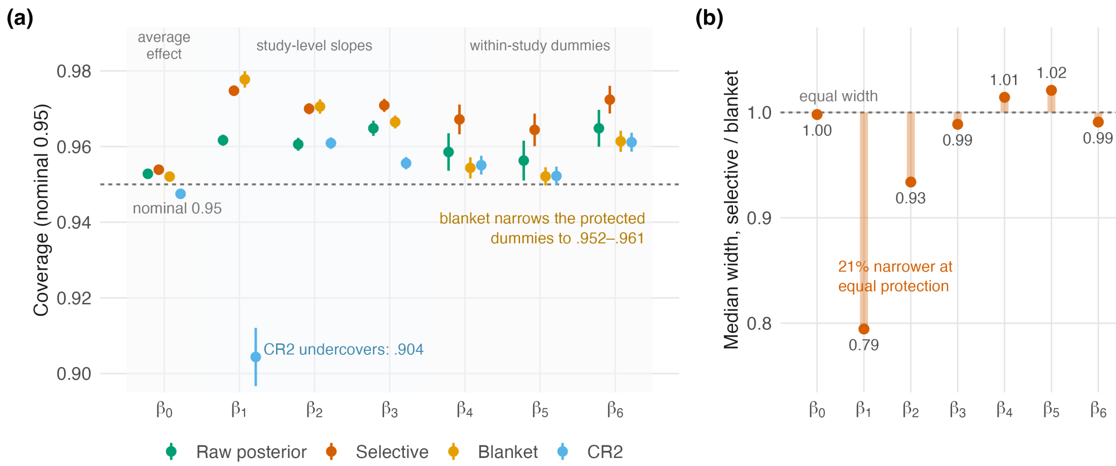

FIGURE 2. (a) Coefficient-level coverage by estimator (correct specification, in-envelope). Points are unweighted means over conditions; bars span ± 1.96 between-condition standard errors; the dashed line marks the nominal .95, and the two headline comparisons are annotated. (b) Median interval width of the selective estimator relative to blanket calibration (the width column of Table 5): equal protection at up to 21% less width on the slopes.

bins, so no conditioning subset behaves anti-conservatively. Note also that the blanket column serves as the comparison in which every coefficient's target variance is matched unconditionally, since this is what blanket calibration does coordinate by coordinate; the raw, selective and blanket columns therefore span no correction, one-sided gating and unconditional matching.

6.2. **Estimand level: the registered-contrast study.** Because the stored simulation retains only marginal quantities, the evaluation of contrasts requires new simulation runs. We generated an 18-cell design (correctly specified CE and CHE arms; $J \in \{16, 24, 40\}$, all within the operating envelope for $K = 7$; $\rho \in \{0, 0.3, 0.8\}$; 500 replications per cell; 99.9% converged), saving the full posterior and target covariance matrices for each replication. Five contrasts were registered before the simulation was run: a within-dummy difference and a within-dummy mean (the analogues of C1 and C2 in Table 2), a study-level moderator difference, and two mixed directions constructed to stress coordinate coupling. Gated intervals use the estimand rescaling (2.1) with the per-estimand Satterthwaite reference distribution, and blanket-projected intervals apply the same scaling unconditionally, providing the comparison in which every estimand is rescaled regardless of its DER. Replications flagged by

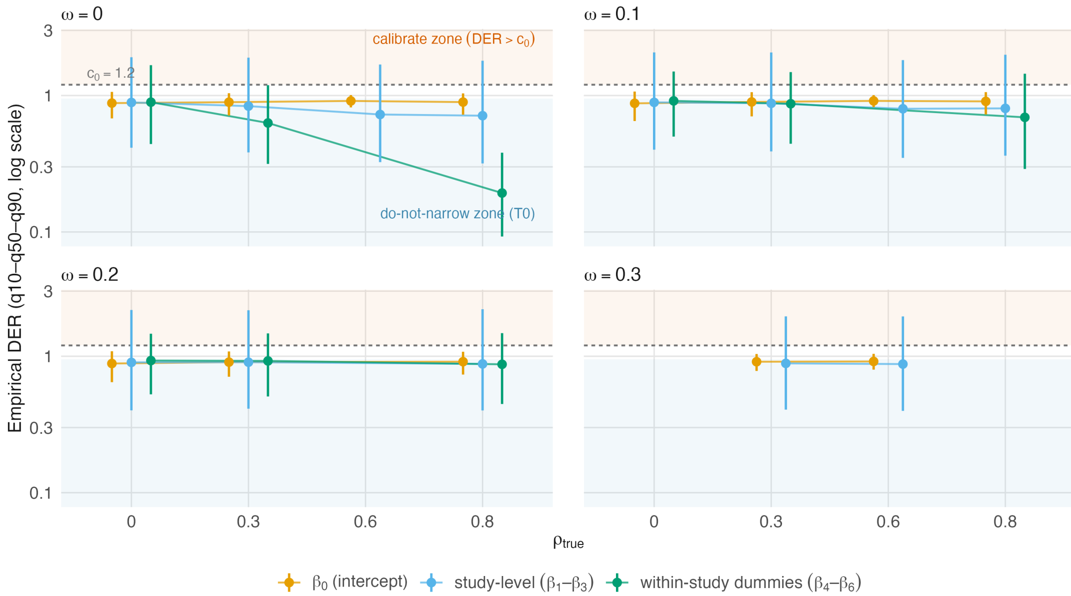

FIGURE 3. Empirical DER distributions (q10–q50–q90, log scale) by coefficient type, true correlation, and heterogeneity. The shaded regions mark the workflow's two one-sided zones: calibration above the gate $c_0 = 1.2$ (dashed) and the do-not-narrow zone below 0.95. Between-study-identified slopes carry the upper tails; within-study dummies concentrate below one.

the fragility rule (mean rates of 0–17% by contrast, reaching 37% in the worst single cell, and concentrated in a near-degenerate moderator direction at small J) are disclosed and excluded for all estimators symmetrically. Fragility is thus itself an estimand-level outcome that the workflow reports: a registered contrast can be practically non-estimable in some designs, and the disclosure rate indicates how often this occurs.

As Table 6 shows, the gated workflow maintains coverage of .970–.975 on all five registered contrasts. It exceeds the raw posterior everywhere, matches blanket calibration where blanket calibration is appropriate (SC3 and SCA2, at 92% of its width), and does better than blanket calibration where the latter narrows protected estimands (SC1, SC2 and SCA1, for which blanket calibration covers at .954–.958). The coefficient-level pattern therefore reproduces at the level of scientific contrasts. The gate itself proves robust in two respects. Gating on the estimated DER loses nothing to a more stable rule: replication-level gating covers .970–.975, compared with .962–.969 for a benchmark rule that gates each cell on its median DER, even though the two rules disagree on 14–27% of replications; the estimated gate adapts to the

TABLE 6. Registered-contrast operating characteristics (nominal 95%; $c_0 = 1.2$). Aggregation conventions: coverage and gate share are unweighted means of the 18 cell-level summaries, computed among non-fragile replications after the disclosure rule of Section 5; the width column is the *median of the 18 cell-level width ratios* (gated over blanket-projected, cell medians); "fragile" is the mean cell-level disclosure rate (full-sample flag rates, which include fragile replications and therefore run higher for SC3/SCA2, are in Supplement S-D). SC1/SC2: within-dummy difference and mean; SC3: study-level difference; SCA1/SCA2: adversarial mixed directions.

Contrast	raw	gated	blanket-projected	share	$\frac{\text{width gated}}{\text{width blanket}}$	fragile (%)
SC1	.962	.972	.958	.18	0.97	4.3
SC2	.969	.975	.954	.17	1.00	1.5
SC3	.962	.973	.973	.27	0.92	16.7
SCA1	.965	.970	.955	.14	1.00	0.0
SCA2	.964	.973	.973	.26	0.92	16.7

realized target. And the flag set is insensitive to the declared working correlation: across $\rho_{\text{work}} = 0.6$ versus 0.8 the flag decisions agree on 100% of replications for every contrast. Coordinate-versus-contrast tier reversals occur at modest per-replication rates in these data-generating processes (0.4–5.3%); the sharper reversals appear in the constructed examples of Table 3 and in the applied analysis of the next section.

7. Application: brief alcohol interventions

We now apply the workflow to the synthesis that anchored the simulations, Tanner-Smith and Lipsey's meta-analysis of brief alcohol interventions for adolescents and young adults (Tanner-Smith & Lipsey, 2015), in the analytic form curated by Pustejovsky and Tipton (2022). The analytic sample comprises $N = 1{,}198$ standardized mean differences from $J = 117$ studies, with between 1 and 108 effect sizes per study (median 6), six outcome categories and five continuous moderators. We fit the CE and CHE working models of Lee and Tipton (2026) with their priors and settings ($\widehat{R}_{\max} \leq 1.009$), declare the operational target at $\rho_{\text{work}} = 0.6$ and evaluate the estimand registry of Table 2.

Table 7 gives the workflow's output for every registered estimand. Seven of the eleven coefficients are flagged, the five moderators and two of the outcome cells, whereas the remaining four cell means are not; among the registered contrasts, C1 is flagged while C2 and C3 are not. Selective calibration therefore widens the seven

TABLE 7. TSL15 audit (CE fit): empirical DER against the declared target; external benchmark (clubSandwich CR2 on a REML refit); theorem-level marginal DER and between weight ω_B (interpretation layer); per-estimand Satterthwaite df. Flag at $c_0 = 1.2$. Theorem-level values at REML $\hat{\tau}^2$, $\rho = 0.6$. †Near-degenerate direction (this moderator is nearly constant across studies): the empirical-vs-external disagreement is a fragility diagnostic, not a stable substantive finding.

Estimand	$\mathrm{DER}_{\mathrm{emp}}$	ext. CR2	thm. marg	ω_B	df	flag
Frequency of use	1.09	1.03	1.20	.81	76	
Frequency of heavy use	1.25	1.27	1.25	.85	76	✓
Quantity of use	1.02	0.99	1.34	.94	93	
Peak consumption	1.24	1.29	1.11	.71	36	✓
Blood alcohol concentration	0.96	1.01	1.08	.67	47	
Combined measures	1.11	1.13	0.98	.58	28	
Follow-up week (within-varying)	2.79	2.98	0.47	.07	6.3	✓
Long follow-up (within-varying)	2.70	2.72	0.42	.02	7.4	✓
% college†	2.73	0.99	1.25	.99	13.4	✓
% male	3.71	6.40	1.21	.83	23.7	✓
Attrition	3.84	4.71	0.81	.43	26.8	✓
C1: heavy − frequency	1.39	1.37	0.41	.01	17.2	✓
C2: grand outcome mean	1.12	1.13	1.37	.96	94	
C3: quantity − BAC	0.92	0.92	0.42	.02	9.7	

flagged coefficients, rescales C1 through its own scalar rule (2.1), and leaves the four unflagged coefficient posteriors, C2, and C3 unchanged. Blanket calibration would have rescaled all of them, narrowing the intervals for blood alcohol concentration and for C3.

The empirical and external columns agree closely for most rows, for instance 1.25 versus 1.27 for heavy use, 2.79 versus 2.98 for follow-up week and 1.39 versus 1.37 for C1; since the two columns come from independent codebases, the agreement is a cross-implementation check. The single sharp disagreement occurs for % college (2.73 versus 0.99), a moderator with almost no variance across these studies. A large disagreement between the declared target and an external benchmark is itself a useful specification diagnostic, and here it identifies the registry's most fragile direction.

Read through ω_B, the theorem-level and empirical layers tell one story. The between-identified estimands, the cell means, % college, % male and the grand mean, all with $\omega_B \geq .58$, fall in the mild range under both layers. The within-identified estimands, the follow-up moderators together with C1 and C3, all with $\omega_B \leq .07$, are protected at the dependence layer, where the theorem-level values are 0.41–0.47, and

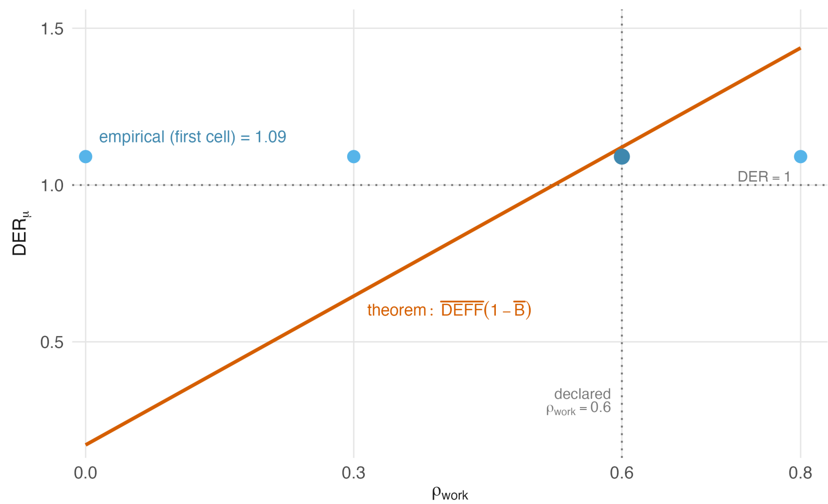

FIGURE 4. TSL15 sensitivity of the intercept-direction DER to the declared working correlation: theorem curve $\overline{\text{DEFF}}(1 - \bar{B})$ with the empirical first-cell overlay; the declared $\rho_{\text{work}} = 0.6$ is marked.

owe their empirical flags to the other components of the declared target. The step-wise decomposition in Supplement S-E identifies the responsible component: moving from the theorem-level target to an uncorrected sandwich with individual-variance weighting at posterior plug-ins already raises the follow-up moderators from 0.47 and 0.42 to 2.35 and 2.37, and C1 from 0.41 to 1.16, with the CR2 adjustment contributing the remainder. The declared weighting convention, rather than the dependence structure alone, drives these values, which is the reason we insist that the target be stated explicitly.

The audit also yields two structural findings. The signed conservation display shows an unbalanced gap of +9.0% at $\rho = 0.6$, attributable entirely to $\widehat{\text{Cov}}(B_j, \text{DEFF}_j)$, as Corollary 1 predicts; Figure 4 traces the full ρ grid. And the CE forest plot requires conditional widening by a median factor of 1.88, with a range of 0.47–8.06 across the 117 studies (Figure 5). This is a study-level consequence of Theorem 3 that no coefficient-level report reveals.

8. Discussion

The companion work of Lee and Tipton (2026) shows how to calibrate a Bayesian posterior to a robust variance target in dependent-effects meta-analysis. In this paper we have considered the questions that precede it: whether the calibration is needed, and for which estimands. Our answer is two-sided. In the conditions that we studied,

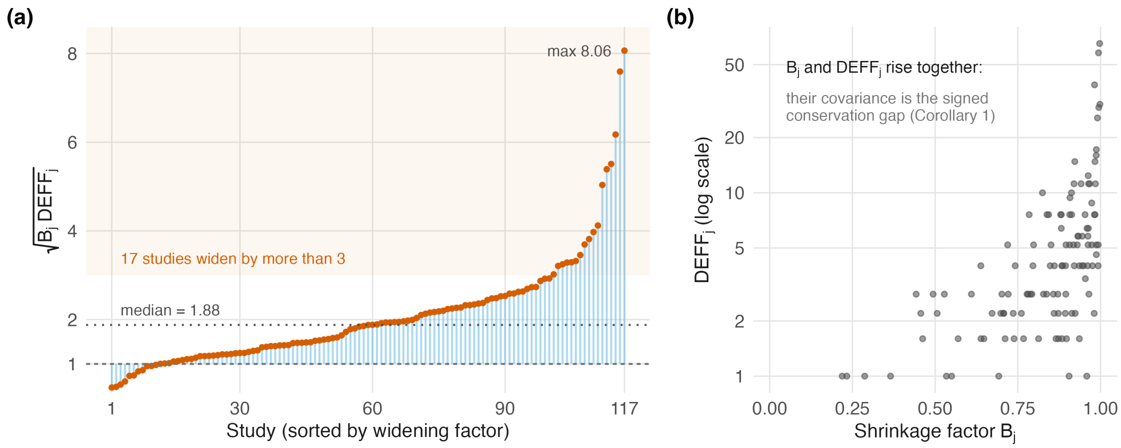

FIGURE 5. TSL15 (CE fit). (a) Per-study conditional widening factors $\sqrt{B_j\,\mathrm{DEFF}_j}$ for forest-plot intervals at $\rho = 0.6$, sorted; median 1.88, and 17 of the 117 studies exceed 3. (b) The mechanism behind the unbalanced conservation gap: B_j and DEFF_j increase together across studies, and their covariance is the signed gap of Corollary 1.

the working posterior is frequently not under-dispersed: raw coverage falls between .953 and .965, median empirical DERs are below one for many coefficients, and applying the correction indiscriminately produces a measurable coverage loss on exactly the estimands that the hierarchical model was protecting. Selective calibration retains the protection afforded by blanket calibration where such protection is needed, at up to a fifth less interval width, and avoids its costs where it is not. The diagnostic that makes this determination is computationally inexpensive, since it uses matrices that the calibrated report already contains, and its decisions proved robust to the choice of gate threshold, to the declared working correlation and to its own estimation noise.

The limitations are worth restating. The decomposition theorems are exact in the hierarchical Gaussian model with known study-average scales and fixed plug-in components, whereas the operational target incorporates further declared components, the individual-variance weighting, the CR2 correction and the plug-in values, whose contribution the TSL15 audit quantifies but the theorems do not model. All coverage statements are specific to the simulation conditions studied, and the fragility rules matter in practice, since near-degenerate moderator directions occur in real syntheses and the workflow discloses such estimands rather than correcting them. Selective correction, finally, is a tool for per-estimand marginal reporting; joint credible regions spanning flagged and unflagged coordinates require the blanket transformation.

Several extensions follow from these limitations. Automated contrast-level software support beyond the computational recipe given here is a natural target for further development. Joint calibrated tests over mixed flag sets remain an open problem, as does the calibration of heterogeneity parameters, in meta-analysis as in the survey setting (Lee, Williams, & Savitsky, 2026). The sampling variability of the diagnostic itself at very low degrees of freedom suggests developing a shrunken or interval-valued version of the DER. Finally, a corpus-level question is now within reach: across published syntheses, how often does each tier occur, and for which types of estimands? Answering this requires only one fitted model per synthesis, and it would provide the design-effect summaries across syntheses that Lee, Tipton, et al. (2026) anticipated as an empirical consequence of the underlying equivalence.

Software and reproducibility. All analyses were conducted in R with posterior sampling via Stan through the `cmdstanr` interface. The workflow runs on the `bayesRVE` package, version 0.7.0 (Lee, 2026) (https://github.com/joonho112/bayesRVE; documentation at https://joonho112.github.io/bayesRVE/), plus a thin analysis module distributed with the replication materials (estimand-level DER tables with tiers, per-estimand Satterthwaite degrees of freedom, fragility flags; selective block calibration; the scalar estimand rescale). Replication is supported at two levels. Verification of every reported number requires only the audit artifacts distributed with the replication package at https://github.com/joonho112/selective-calibration-replication (companion guide at https://joonho112.github.io/selective-calibration-replication/): the frozen aggregate tables (CSV), the analysis module and scripts, and the fitted-model summaries. A full independent rerun additionally requires the repository's stored simulation output (the 5.7-million-row per-replication table and the per-replication covariance shards of the registered-contrast grid) and, if the simulations are to be regenerated from scratch, re-execution of the MCMC designs specified in Supplement S-D; the supplement states which artifact each table depends on.

Data availability. The empirical analysis uses the Tanner-Smith and Lipsey synthesis of brief alcohol interventions (Tanner-Smith & Lipsey, 2015) in the analytic form curated and made available by Pustejovsky and Tipton (2022); the frozen analytic sample (1,198 effect sizes, 117 studies) and its construction are specified in Supplement S-E.

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Online Supplement

Which Coefficients Need Correction? Design-Effect-Ratio Diagnostics and Selective Calibration for Bayesian Meta-Analysis With Dependent Effect Sizes

JoonHo Lee

Contents

Appendix S-A. Conditions, targets, and proofs

S-A.1. **Setting and conditions.** Studies $j = 1, \ldots, J$ contribute n_j estimates; the CE working model is $y_{ij} = x_{ij}^\top \beta + \theta_j + \varepsilon_{ij}$, $\theta_j \sim N(0, \tau^2)$, with working within-study independence at the study-average scale $\bar{s}_j^2$ given θ_j, so the working marginal covariance is $\Sigma_{w,j} = \bar{s}_j^2 I_{n_j} + \tau^2 \mathbf{1}\mathbf{1}^\top$. True sampling errors are equicorrelated, $\mathrm{Cov}(\varepsilon_j) =$

$\bar{s}_j^2 C_{n_j}(\rho)$ with $C_n(\rho) = (1-\rho)I_n + \rho\mathbf{1}\mathbf{1}^\top$. Define

$$a_j = n_j/\bar{s}_j^2, \qquad B_j = \frac{\tau^2}{\tau^2 + \bar{s}_j^2/n_j}, \qquad \mathrm{DEFF}_j = 1 + (n_j - 1)\rho,$$
$$\bar{x}_j = n_j^{-1} X_j^\top \mathbf{1}, \qquad W_j = \bar{s}_j^{-2}\, \tilde{X}_j^\top \tilde{X}_j,$$

with $\tilde{X}_j$ the within-study centered design; note $\tau^2 a_j = B_j/(1-B_j)$.

Conditions.

- (M1) CE working model above with known $\bar{s}_j^2$.
- (M2) $\tau^2 > 0$ fixed (plug-in); propagation of τ^2 uncertainty is the marginal-frame refinement of Lee, Tipton, et al. (2026) and is not modeled here.
- (M3) Flat (or negligible-precision) prior on β. Quantified negligibility: on the TSL15 design, replacing flat with $N(0, 10^2)$ shifts every registry DER by at most 1.5×10^{-4} (relative).
- (M4) True errors equicorrelated with $\rho \in [0, 1)$ within studies, independent across studies.
- (M5) $\sum_j X_j^\top X_j$ full rank; with (M2) all bread matrices below are positive definite.

S-A.2. **Bread, meats, and the posterior identity.** The working score of β from study j is $t_j = X_j^\top \Sigma_{w,j}^{-1} r_j$. From $\Sigma_{w,j}^{-1} = \bar{s}_j^{-2}\{I - (B_j/n_j)\mathbf{1}\mathbf{1}^\top\}$,

$$t_j = \bar{s}_j^{-2}\big\{\tilde{X}_j + (1-B_j)\,\mathbf{1}\bar{x}_j^\top\big\}^\top r_j, \qquad \mathbf{S} = \sum_j X_j^\top \Sigma_{w,j}^{-1} X_j = \sum_j \big\{W_j + a_j(1-B_j)\bar{x}_j\bar{x}_j^\top\big\}. \tag{S-A.1}$$

Under (M3) and fixed τ^2, Gaussian conjugacy gives $\beta \mid y \sim N(\hat{\beta}, \mathbf{S}^{-1})$ exactly, so the DER denominator is $c^\top \mathbf{S}^{-1} c$.

Two meat matrices are required, one for each of the two named targets. Conditional on θ (dependence layer only), $\mathrm{Cov}(r_j \mid \theta) = \bar{s}_j^2 C_{n_j}(\rho)$; marginally, add $\tau^2\mathbf{1}\mathbf{1}^\top$. Using $C_n(\rho)\mathbf{1} = \mathrm{DEFF}\,\mathbf{1}$ and $\tilde{X}_j^\top \mathbf{1} = 0$ (cross terms vanish),

$$\mathbf{M}_{\mathrm{cond}} = \sum_j \Big\{(1-\rho)W_j + \mathrm{DEFF}_j(1-B_j)^2 a_j\, \bar{x}_j\bar{x}_j^\top\Big\}, \tag{S-A.2}$$

$$\mathbf{M}_{\mathrm{marg}} = \sum_j \Big\{(1-\rho)W_j + a_j(1-B_j)\big[(1-B_j)\mathrm{DEFF}_j + B_j\big]\bar{x}_j\bar{x}_j^\top\Big\}, \tag{S-A.3}$$

where (S-A.3) uses $\tau^2 a_j = B_j/(1-B_j)$. Both displays, and (S-A.1), are verified against direct $X^\top \Sigma_w^{-1}(\cdot)\Sigma_w^{-1} X$ computation to relative Frobenius error $\leq 2.7 \times 10^{-16}$ and against Monte Carlo score covariances at the expected $O(R^{-1/2})$ (Section S-G).

Remark S-A.1 (Joint-system identity)**.** The β-block of the joint (β,θ) sandwich, with the posterior Hessian including the prior precision $1/\tau^2$ on θ as bread and the likelihood scores as meat, equals the profiled $\mathbf{S}^{-1}\mathbf{M}\mathbf{S}^{-1}$ exactly, by the block-inverse identity applied to both bread and (profiled) scores; verified numerically to 2.9×10^{-15}. Population statements here are therefore invariant to which of the two equivalent constructions is used.

S-A.3. **Proof of Theorem 1 (directional decomposition).** The general expression follows from Definition 1 applied to (S-A.1)–(S-A.2). For part (a), consider the between direction and drop W_j (study-level covariates have $\tilde{X}_j$-columns zero) and set $n_j \equiv n$, $\bar{s}_j^2 \equiv \bar{s}^2$: the intercept-only scalars are $S_\mu = \sum_j a(1-B)$ and $M_\mu = \sum_j \mathrm{DEFF}(1-B)^2 a$, so $\mathrm{DER} = \mathrm{DEFF}(1-B)$; with (S-A.3), $M_\mu^{\mathrm{marg}} = \sum_j a(1-B)[(1-B)\mathrm{DEFF}+B]$ gives $B+\mathrm{DEFF}(1-B)$. For part (b), if $\bar{x}_j^\top \mathbf{S}^{-1}c = 0$ for all j, then with $u = \mathbf{S}^{-1}c$ all between terms vanish from both quadratic forms, leaving $u^\top \mathbf{M}_{\mathrm{cond}} u = (1-\rho)\, u^\top(\sum_j W_j)u = (1-\rho)\, u^\top \mathbf{S} u$. □

Remark S-A.2 (The representer condition)**.** The hypothesis of part (b) concerns the precision-weighted representer $u = \mathbf{S}^{-1}c$ and not c itself. A within-study-centered covariate that is correlated within studies with other moderators couples the coordinates through the off-diagonal elements of W_j, and its coordinate DER then deviates from $1-\rho$; the exact value in that case is given by the mixture representation of Theorem 2. The hypothesis holds exactly for a within-centered covariate that is orthogonal, within studies, to all other columns, and the two-coefficient model consisting of an intercept and such a covariate satisfies it identically.

S-A.4. **Proof of Theorem 2 (mixture representation and $R(c)$ bounds).** Write $\mathrm{DER}_{\mathrm{cond}}(c^\top\beta) = u^\top \mathbf{M}_{\mathrm{cond}} u / u^\top \mathbf{S} u$ with $u = \mathbf{S}^{-1}c$, and expand numerator and denominator over (S-A.1) and (S-A.2):

$$u^\top \mathbf{S} u = q_W + q_B, \quad q_W = u^\top\Big(\sum_j W_j\Big)u, \quad q_B = \sum_j (1-B_j)a_j(\bar{x}_j^\top u)^2,$$

$$u^\top \mathbf{M}_{\mathrm{cond}} u = (1-\rho)\, q_W + \sum_j \mathrm{DEFF}_j(1-B_j)^2 a_j(\bar{x}_j^\top u)^2.$$

Dividing the two expansions and identifying $\omega_B(c) = q_B/(q_W+q_B)$ and $\overline{\mathrm{DEFF}}_B(c)$ as the ratio of the between sums. When $q_B = 0$ (a purely within-identified estimand) the representation reduces to its within term: $\omega_B(c) = 0$, the between term is absent, and $\overline{\mathrm{DEFF}}_B(c)$ is left undefined by convention, as stated explicitly in the theorem. For the bounds: $\mathbf{H}_{\beta\beta} = \sum_j (W_j + a_j\bar{x}_j\bar{x}_j^\top)$ satisfies $\mathbf{S} = \mathbf{H}_{\beta\beta} - \sum_j B_j a_j \bar{x}_j\bar{x}_j^\top \preceq \mathbf{H}_{\beta\beta}$, hence $c^\top \mathbf{S}^{-1} c \ge c^\top \mathbf{H}_{\beta\beta}^{-1} c$ and $R(c) \ge 0$; and $\sum_j B_j a_j \bar{x}_j \bar{x}_j^\top \preceq B_{\max} \mathbf{H}_{\beta\beta}$ gives $\mathbf{S} \succeq (1 - B_{\max}) \mathbf{H}_{\beta\beta}$, hence $c^\top \mathbf{S}^{-1} c \le (1 - B_{\max})^{-1} c^\top \mathbf{H}_{\beta\beta}^{-1} c$ and $R(c) \le B_{\max}$. $\square$

Remark S-A.3 (Reparameterization)**.** Under $\beta = A\alpha$, scores map by $A^\top$; hence $\mathbf{S}_\alpha = A^\top \mathbf{S} A$, $\mathbf{M}_\alpha = A^\top \mathbf{M} A$, and for $d = A^\top c$, $u_\alpha = \mathbf{S}_\alpha^{-1} d = A^{-1} u$, so that both quadratic forms, and every ingredient of the mixture, are invariant for the fixed estimand. Verified on random nonsingular A (toy) and on the cell-means-to-reference-cell recoding of the TSL15 design (4×10^{-16}).

S-A.5. **Proof of Theorem 3 and Corollary 1.** For θ_j at fixed β: bread $a_j + 1/\tau^2$; conditional meat $\mathrm{Var}(\bar{s}_j^{-2} \mathbf{1}^\top \varepsilon_j) = \bar{s}_j^{-4} \cdot \bar{s}_j^2 \, \mathbf{1}^\top C \mathbf{1} = a_j \mathrm{DEFF}_j$; posterior variance $(a_j + 1/\tau^2)^{-1} = B_j / a_j$. The sandwich variance is $a_j \mathrm{DEFF}_j / (a_j + 1/\tau^2)^2$, and the ratio to the posterior variance is $B_j \mathrm{DEFF}_j$. The finite-J factor $\kappa_j(J)$ arises from the coupling of θ_j to the grand mean under a flat prior on μ in the balanced intercept-only model; we import the survey-side derivation (Lee, Williams, & Savitsky, 2026) through the condition map of Section S-B and verify it here by direct computation of the joint $(K{+}J)$-dimensional sandwich (maximum relative deviation 1.9×10^{-15} across all studies of a $J = 15$ balanced design). Conservation and the boundary rows of Corollary 1 follow by substitution; the unbalanced signed gap is the algebraic identity $\overline{B_j \mathrm{DEFF}_j} - \bar{B}\, \overline{\mathrm{DEFF}} = \widehat{\mathrm{Cov}}(B_j, \mathrm{DEFF}_j)$. $\square$

Appendix S-B. CHE substitution and the survey-to-meta condition map

Proposition S-B.1 (Two-parameter CHE substitution)**.** *Under the CHE working model with within-study heterogeneity ω^2 (conditional on plug-in ω^2), all objects of Section S-A are the CE objects after*

$$\bar{s}_j^2 \;\mapsto\; \bar{s}_j^2 + \omega^2, \qquad \rho \;\mapsto\; \rho_j^{\mathrm{che}} = \frac{\rho\, \bar{s}_j^2}{\bar{s}_j^2 + \omega^2},$$

i.e., the total conditional error covariance is $(\bar{s}_j^2 + \omega^2)\, C_{n_j}(\rho_j^{\mathrm{che}})$. Both parameters transform, and the equicorrelation is diluted study by study. The identity is verified numerically to 3×10^{-16} for both the bread and the meat.

Because the CHE implementation integrates θ_j out analytically, no CHE random-effect DER exists as a software object; all CHE statements in the paper are β-block and plug-in-conditional.

Two items of the framework's original (v1) statement do not appear in the table because they are no longer conditions on the survey side itself. The design central limit theorem, formerly condition A5, is now the named target-validity condition

TABLE S-B.1. Survey-side conditions of Lee, Williams, and Savitsky (2026) and their meta-analytic disposition. The survey-side inventory is that of the current version of the framework: design and weight conditions (A1)–(A4), model-structure conditions (R1)–(R3) and (R5), and the named target-validity condition (TV).

Survey condition	Disposition	Replacement / anchor
A1 informative sampling, weights	dropped	unit weights (bridge boundary, Lee, Tipton, et al., 2026)
A2 pseudo-likelihood regularity	dropped	exact Gaussian algebra at fixed τ^2
A3 prior support	(M3)	flat/negligible β-prior, quantified
A4 weight boundedness	dropped	unit weights
R1 proper RE prior	(M2), (M5)	$\tau^2 > 0$
R2 group–PSU nesting	automatic	study = cluster
R3 flat prior on fixed effects	(M3)	as above
R5 Schur non-degeneracy	(M5)	full-rank design, $\tau^2 > 0$
TV target validity	replaced	study-level CLT, $J \to \infty$ (Lee, Tipton, et al., 2026)

(TV), which records the imported asymptotic meaning of the declared target; its meta-analytic analogue is the study-level CLT anchoring of Lee, Tipton, et al. (2026), as the final row states. The scalar within-group design-effect condition, formerly R4, was deleted there: under the conjugate Gaussian scope the scalar within-group form of the meat is the parameterization of the survey-side theorems rather than an assumption about the design, with heterogeneous design effects read as score-weighted within-group averages. The meta-analytic development sharpens the same point from the other side. The explicit equicorrelation supplies the meat directly, and the two-eigenvalue structure of $C_n(\rho)$ resolves the within and between directions exactly (Theorem 1), so no scalar parameterization arises at any step.

Appendix S-C. Per-estimand Satterthwaite df and fragility rules

S-C.1. **Per-estimand degrees of freedom.** The CR2 construction supplies, per study, the adjustment matrix A_j, weight matrix W_j, design block X_j, and bread $Q = (\sum_j X_j^\top W_j X_j)^{-1}$. The Bell–McCaffrey adjustment and the Satterthwaite two-moment approximation are formulated for arbitrary linear combinations $c^\top \hat{\beta}$, not only coordinates (Bell & McCaffrey, 2002; Satterthwaite, 1946; Tipton, 2015); the construction below is that formulation at the estimand's own projection vectors. For an estimand c, define $g_j(c) = A_j^\top W_j X_j Q\, c$ and the $J \times J$ matrix

$$P_{ii} = g_i^\top W_i^{-1} g_i - g_i^\top X_i Q X_i^\top g_i, \qquad P_{i\ell} = -\,(X_i^\top g_i)^\top Q\,(X_\ell^\top g_\ell) \;\; (i \neq \ell),$$

and $\mathrm{df}(c) = \{\mathrm{tr}(P)\}^2/\,\mathrm{tr}(P^2)$: the two-moment Satterthwaite approximation of Satterthwaite (1946) and Tipton (2015), evaluated at the estimand's own projection vectors. At $c = e_k$ this reproduces the per-coefficient implementation of Lee and Tipton (2026) to 3.7×10^{-16} (relative, all TSL15 coefficients). For the TSL15 contrasts, the resulting values are 17.2 (C1), 94.0 (C2), and 9.7 (C3).

S-C.2. **Fragility rules.** A DER is reported only if all four of the following conditions hold: (F1) estimability, that is c in the column space of $\mathbf{S}$ (relative reconstruction residual $< 10^{-6}$); (F2) $c^\top V_T c \geq 0$ (the quadratic form, not global definiteness); (F3) $c^\top \Sigma c \geq 10^{-8}\,\mathrm{tr}(\Sigma)/K$; (F4) Monte Carlo adequacy, $\mathrm{MCSE}\{\log \widehat{\mathrm{DER}}\} \approx \sqrt{2/S_{\mathrm{eff}}} < 0.05$. For the registered-contrast grid we additionally operationalize a disclosure rule ($\mathrm{DER} > 50$ or $\mathrm{df}(c) < 2$), which fired at mean rates of 0–17% by contrast, reaching 36.6% in the worst single cell (SC3 at $J = 16$), and was concentrated where the resampled %-college covariate is nearly constant at small J: a direction that is nearly non-estimable, and which is disclosed for all estimators symmetrically (Table S-D.2). Fragility is reported as a third workflow outcome rather than as a mere exclusion. Degrees of freedom just above the hard cut, that is $2 \leq \mathrm{df}(c) < 6$, do not lead to exclusion but do trigger the unstable-gate label of the main text (Section 5).

Appendix S-D. Simulation details

S-D.1. **Coefficient-level re-analysis (no new sampling).** Source: the stored replication layer of Lee and Tipton (2026) (5,745,500 rows; one row per condition $\times$ replication $\times$ method $\times$ coefficient). Methods used: the calibrated CCC-t estimators (CE and CHE arms), their uncalibrated raw-Bayes partners sharing the same Stan fits, and the frequentist CR2 benchmarks. Filters, in order: correct specification (CE method under CE generation; CHE under CHE), operating envelope $J \geq \max(8, 2K+2)$, joint convergence of the three comparators. Analysis set: 955,762 rows.

Identity verification. The stored calibration ratio r_k equals $\sqrt{\mathrm{DER}}$ against the CR2 target by the exact-matching property; we verified the identity before use by comparing r_k^2 with $(\mathrm{se}_{\mathrm{cal}}/\mathrm{se}_{\mathrm{raw}})^2$ row by row: 1,904,177 usable rows; median relative deviation 2.7×10^{-16} (CE) and 0 (CHE); 99th percentile $\leq 4.9 \times 10^{-15}$; no row exceeded 10^{-2}. The 3.1% of joined rows excluded were exactly the non-converged replications.

Selective synthesis. Per replication and coefficient: flagged $\Leftrightarrow r_k^2 > c_0$; the selective interval is the calibrated interval if flagged, the raw posterior interval otherwise. Aggregation: per-condition means, then unweighted means over conditions with between-condition standard errors; binomial MCSEs ≤ 0.001 throughout.

TABLE S-D.1. Threshold sweep (selective estimator; correct specification, in-envelope). Left: small slope β_1 (coefficient level). Right: registered contrasts (grid, pooled).

	β_1 (re-analysis)		contrasts (grid)		
c_0	coverage	median width	coverage	width ratio	gate share
1.05	.977	.669	.975	.990	.29
1.10	.976	.662	.974	.977	.26
1.20	.975	.650	.973	.964	.21
1.50	.972	.623	.969	.936	.10
2.00	.969	.596	.966	.921	.04

Gate noise. Under exact correct specification ($\rho = 0$, $\omega = 0$), per-replication DER quantiles (q10–q90) span 0.68–1.06 for the average effect but 0.41–1.91 for slopes; flag rates at $c_0 = 1.2$ are 3.5% and 27.5% respectively. The wide slope spread reflects sandwich estimation noise at effective degrees of freedom near 4–6.

Conditional-coverage diagnostics. To address the concern that gating on an estimated quantity could induce selection effects, we computed, from the same stored data, coverage of the gated report conditional on the gate's own behavior (nominal 95%; correct specification, in-envelope):

Conditioning	Bin	n	coverage	median width
gate status	unflagged	782,253	.959	0.32
	flagged	173,517	.995	1.29
threshold proximity	$\lvert\log(\mathrm{DER}/c_0)\rvert \leq 0.1$	92,590	.970	0.48
	$(0.1, 0.25]$	197,495	.962	0.24
df (flagged only)	df ≤ 4	50,077	1.000	3.04
	$(4, 6]$	32,528	.998	1.78
	$(6, 10]$	42,636	.996	1.08
	$(10, 20]$	35,636	.991	0.66
	> 20	12,640	.982	0.19
DER bin (extremes)	$(0.95, 1.05]$	109,597	.954	0.18
	$(4, \infty)$	30,171	1.000	—

No conditioning subset exhibits anti-conservative behavior. Coverage is lowest (.954–.962) in the calibrated mid-bins where raw and calibrated intervals nearly coincide, .970 in the nearest threshold band, and *increases* as flagged degrees of freedom fall, which is the direction opposite to what a selection effect would produce. The $(4, \infty)$ DER bin is dominated by the same near-degenerate directions that the fragility rule discloses, and its widths are therefore not representative.

S-D.2. **Registered-contrast grid.** Design (frozen before launch; the registry addendum is in the repository): $K = 7$ (intercept; three study-level moderators; three within-study outcome-category dummies); correctly specified CE ($\omega = 0$) and CHE ($\omega = 0.1$) arms; $J \in \{16, 24, 40\}$ (all within the envelope $J \geq 2K + 2 = 16$); $\rho \in \{0, 0.3, 0.8\}$; $\tau = 0.2$; cluster sizes and moderators resampled from the TSL15 database; 500 replications per cell; deterministic seeds; convergence screen $\widehat{R}_{\max} < 1.05$ (8,995 of 9,000 replications retained). Stored per replication: full Σ, V_T at both declared $\rho_{\text{work}} \in \{0.8, 0.6\}$, and per-contrast sufficient statistics (center, raw sd, raw quantile interval, DER and $\mathrm{df}(c)$ at both targets), from which gated and blanket-projected intervals reconstruct for any c_0.

Registered contrasts and true values, in the coefficient basis $(\beta_0, \ldots, \beta_6)$: SC1 = $e_5 - e_6$ (within-dummy difference; true 0.0190); SC2 = $(e_5+e_6+e_7)/3$ (true 0.0135); SC3 = $e_2 - e_3$ (study-level difference; true -0.2603); SCA1 = $e_1 - (e_5+e_6+e_7)/3$ (true 0.1270); SCA2 = $e_2 - e_3 + e_5 - e_6$ (true -0.2413).

Table S-D.2. Grid diagnostics by contrast (pooled over 18 cells). Flag rates at $c_0 = 1.2$; dual-target agreement between $\rho_{\text{work}} = 0.8$ and 0.6; stability = coverage of estimated-gate vs. cell-median-gate; reversals = per-replication tier disagreements between the contrast and its component coefficients; fragility = disclosure rate.

	flag rate	dual agree	cov. est-gate	cov. med-gate	reversal (%)	fragility
SC1	.183	1.00	.972	.962	5.3 / 3.0	.043
SC2	.168	1.00	.975	.969	0.4 / 0.1	.015
SC3	.388	1.00	.973	.962	0.9 / 0.6	.167
SCA1	.138	1.00	.970	.965	0.7 / 0.0	.000
SCA2	.382	1.00	.973	.964	0.6 / 0.1	.167

Appendix S-E. TSL15 audit details

Sample accounting (frozen ledger). Raw curated subset: 1,289 rows, 133 study identifiers. Complete cases on the eight analysis columns (effect size, variance, study and estimate identifiers, outcome category, % male, % college, attrition): $N = 1{,}198$ and $J = 117$, reproducible from the raw file and matching the curators' own description (117 studies; 1–108 effect sizes per study, median 6, IQR 3–12; Pustejovsky and Tipton, 2022). Six outcome categories with 56–470 estimates each; 94 of 117 studies contribute to more than one category. All fits cluster on the study identifier ($J = 117$).

Model and settings. Cell-means meta-regression on outcome category plus five moderators ($K = 11$); priors $\beta_k \sim N(0, 10^2)$, half-Cauchy$(0, 0.5)$ on τ (and ω, CHE); four chains, 2,000 warmup and 2,000 sampling ($S = 8{,}000$), fixed seed; $\widehat{R}_{\max} = 1.009$ (CE), 1.001 (CHE). Declared operational target: CR2-corrected marginal sandwich at $\rho_{\text{work}} = 0.6$ and posterior-mean plug-ins. External benchmark: REML refit with imputed $\rho = 0.6$ covariances and clubSandwich CR2. Theorem layer: REML $\hat{\tau}^2 = 0.0431$ on the same design matrices.

Signed conservation display ($\widehat{\text{Cov}}$ = empirical covariance across the 117 studies):

ρ	$\overline{\text{DEFF}}$	signed gap	gap (%)	$\widehat{\text{Cov}}(B_j, \text{DEFF}_j)$
0.0	1.00	0.000	0.0	0.000
0.3	3.77	+0.294	+7.8	0.294
0.6	6.54	+0.588	+9.0	0.588
0.8	8.39	+0.784	+9.3	0.784

The gap equals the covariance term identically, as Corollary 1 states; the balanced finite-J deficit is negligible at $J = 117$.

Target-layer decomposition (the theorem-to-operational gap, quantified). For each registry estimand, the DER is recomputed under a ladder of targets on the same fitted model: L0 = theorem-layer marginal (study-average scales, flat prior, REML $\hat{\tau}^2$); L1 = uncorrected empirical sandwich with individual-variance weighting at posterior plug-ins (`small_sample = "none"`); L2 = the declared CR2 target; L3 = the external REML/clubSandwich benchmark. Selected rows are shown below.

Estimand	L0 theorem	L1 per-obs HC0	L2 CR2 (declared)	L3 external
Follow-up week	0.47	2.35	2.79	2.98
Long follow-up	0.42	2.37	2.70	2.72
% college†	1.25	3.04	2.73	0.99
% male	1.21	4.42	3.71	6.40
Attrition	0.81	3.83	3.84	4.71
C1: heavy − frequency	0.41	1.16	1.39	1.37
C3: quantity − BAC	0.42	0.82	0.92	0.92
Frequency of heavy use (cell)	1.25	1.17	1.25	1.27

The pattern is consistent across rows: every large difference between the theorem-level and operational values arises at the L0→L1 step, in which study-average scales and the flat-prior profile are replaced by individual-variance weighting at posterior plug-ins, while the CR2 adjustment (L1→L2) contributes a smaller further change and the external benchmark agrees with L2 except in the near-degenerate %-college direction. This decomposition quantifies the role of the declaration: the weighting convention, rather than the dependence structure or the small-sample correction, is the component that re-exposes the within-varying moderators and C1 in these data. (Full table: `rc12_layer_decomposition.csv`; the L2 column reproduces the frozen audit values to 4×10^{-15}.)

Selective impact and forest. At $c_0 = 1.2$ the flagged set is the seven coefficients marked in the audit table of the main text (Table 7); the selective report calibrates those (block-Cholesky on F, per-coefficient Satterthwaite t) and retains raw posterior quantile intervals for the four unflagged cell means. Study-effect intervals (CE fit) widen by $\sqrt{B_j \mathrm{DEFF}_j}$ with median 1.88 and range 0.47–8.06; 17 of 117 studies have factors above 3.

Appendix S-F. Software

The workflow runs on `bayesRVE` 0.7.0 (Lee, 2026) plus a thin analysis module distributed with the replication materials: `der_table()` (estimand-level DER with tiers, per-estimand df, and fragility flags), `ccc_selective()` (block-Cholesky on a flagged set; identical to the package's blanket transform when F is the full coefficient set, verified byte-exactly), `conservation_signed()` (signed display with the covariance decomposition), and the scalar estimand rescaling of the main text. Acceptance tests (13 checks: hand-parity of every DER; per-coefficient df parity; blanket/empty/mixed-F semantics; flag firing and silence; conservation identity; rescale exactness) all pass at machine precision; the test script and outputs are in the repository. As noted in the main text, `ccc_selective()` matches the target on the flagged coordinates but alters every linear combination inside the block; registered contrasts must be computed through `rescale_contrast()` (their own scalar rule), never by projecting block-transformed draws.

Appendix S-G. Verification ledger

Suite	Checks	Result	Worst deviation
Constructed-truth algebra + MC (bread, meats, mixture, invariance, bounds, boundary, conservation, κ_j, joint block, CHE)	18	all pass	6×10^{-16} alg.; MC ≈ 0.03 at R=6000
TSL15 real-design checks (mixture, bounds, recoding invariance, prior negligibility, $K = 11$)	5	all pass	1.6×10^{-14}
Stored-simulation identity (r_k^2 vs. ratio route, 1.9M rows)	1	pass	p99 $\leq 4.9 \times 10^{-15}$
Software acceptance (module vs. package, flags, semantics)	13	all pass	byte-exact parity

We record two corrections that emerged from the verification process itself: the reparameterization rule was first implemented with the covariance transform applied to precision matrices (caught by a 0.62 relative deviation and corrected to $A^\top(\cdot)A$); and the within-direction result was first stated without the representer condition (caught by a 10^{-7}-level deviation in a coupled design and restated as in Section S-A).